BASIL C K

AI ENGINEER | INDUSTRIAL AI & AUTOMATION | COMPUTER VISION | GENERATIVE AI

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PROFILE

AI Engineer at Difinity Digital LLP with hands-on experience developing **AI-driven industrial automation, computer vision, machine learning, and Generative AI solutions**. Experienced in building industrial monitoring pipelines using YOLOv26 detection and segmentation models, integrating the Segment Anything Model (SAM) into annotation workflows, and applying **BERT-based Transformer models** to application-specific NLP tasks. Skilled in Python backend development, LLM integration, Retrieval-Augmented Generation (RAG), vector databases, real-time streaming, and AI workflow orchestration. Developing OpenMind AI Platform for multi-provider LLM evaluation and orchestration. Author of peer-reviewed research published in **IEEE Xplore** on hybrid CNN–YOLO architecture for vehicle damage assessment.

TECHNICAL SKILLS

Machine Learning & Computer Vision: YOLOv26 (Nano/Small/Medium/Large), YOLOv26-Seg, SAM, CNNs, BERT, Object Detection, Instance Segmentation, Image Classification, Model Training & Evaluation, Industrial Monitoring & Automation, Object Counting, Conveyor Belt Monitoring

LLM & Generative AI: LLM APIs, RAG, Prompt Engineering, Multi-Provider Inference, LLM Evaluation, Embeddings, Semantic Search, SSE Streaming, AI Memory Systems, Agentic AI, Tool Calling, Workflow Orchestration

Frameworks & Libraries: PyTorch, TensorFlow/Keras, OpenCV, Ultralytics, FastAPI, Flask, SQLAlchemy, Pydantic, NumPy, Pandas, Label Studio

Backend, Infrastructure & Tools: Python, SQL, REST APIs, PostgreSQL, pgvector, Docker, Docker Compose, Git, Linux, VS Code, Jupyter Notebook

Cloud & Certifications: Oracle Cloud Infrastructure (AI Foundations Associate), AWS Fundamentals, Google AI Essentials

EXPERIENCE

AI Engineer — Associate 1
Sep 2026 – Present

Difinity Digital LLP *Kochi, Kerala*

- Develop AI-driven solutions for **industrial automation** and application-specific use cases, applying **computer vision**, **machine learning**, and **BERT**-based Transformer models to visual monitoring and NLP tasks.
- Build and maintain **YOLOv26**-based **object detection** and **segmentation** pipelines for industrial monitoring, object localization, and automated visual analysis.
- Develop **Generative AI** applications integrating multiple LLM providers, including OpenAI, Google Gemini, and NVIDIA, with configurable inference and real-time response streaming.
- Develop **RAG** pipelines involving document ingestion, text chunking, embedding integration, and pgvector-based semantic retrieval for context-aware AI applications.
- Design multi-step **agentic AI** workflows incorporating conditional branching, tool use, and workflow orchestration for complex automation tasks.
- Build Python backend services and REST APIs using FastAPI, with PostgreSQL, pgvector, and Docker-based development environments.
- Collaborate with engineering teams on system architecture, implementation, code reviews, and sprint planning.

AI Engineer Intern — Difinity Digital
Jun 2026 – Sep 2026

Difinity Digital LLP *Kochi, Kerala*

- Developed **YOLOv26** object detection and segmentation pipelines for industrial monitoring use cases, including sack detection, conveyor belt tracking, static object monitoring, and zone-based activity detection.

- Benchmarked YOLOv26 Nano, Small, Medium, and Large variants, evaluating accuracy-latency trade-offs to identify suitable deployment configurations.
- Integrated Meta's Segment Anything Model (**SAM**) into Label Studio through a custom backend, enabling one-click segmentation annotation and streamlining data labeling workflows.

PROJECTS

OpenMind AI Platform — LLM Evaluation & Orchestration Platform ONGOING

Jul 2026 – Present

Python | FastAPI | LLM APIs | RAG | pgvector | SQLAlchemy | Docker | PostgreSQL

- Developed a multi-session AI playground for evaluating LLM responses with configurable inference parameters, including temperature, top_p, max_tokens, and system prompts.
- Implemented real-time SSE streaming for interactive model evaluation and engineered an OpenAI-compatible API gateway with Bearer token authentication, structured logging, and monitoring across 40+ standardized API endpoints.
- Developing a **RAG** pipeline for PDF and Markdown ingestion, text chunking, embedding integration, and pgvector-based semantic retrieval to evaluate different embedding and retrieval configurations.
- Designing a graph-based memory architecture covering conversational, semantic, and long-term memory, alongside multi-step orchestration for LLM routing, conditional branching, and agentic workflows.

Hybrid CNN–YOLO Framework for Car Damage Assessment

2025 – 2026

Python | TensorFlow/Keras | YOLO | OpenCV | FastAPI | Computer Vision **(Published)**

- Developed a **hybrid deep learning** framework combining **YOLO**-based object detection with **CNN**-based severity classification for automated vehicle damage localization and severity assessment.
- Built an end-to-end machine learning pipeline covering dataset preparation, model training, hyperparameter optimization, and evaluation using mAP, F1-score, precision, and recall.
- Developed a FastAPI inference backend to support automated vehicle damage predictions.

PUBLICATIONS

A Hybrid CNN–YOLO Framework for Car Damage Severity Classification and Localization

2026 IEEE International Conference on Natural Language Processing and Computer Vision (ICNPCV), Bengaluru, India

- Published in **IEEE Xplore**, featuring peer-reviewed research on deep learning for automated vehicle damage assessment.
- Proposed a hybrid YOLO–CNN framework for vehicle damage localization and severity classification.

EDUCATION

B.Tech in Artificial Intelligence and Data Science

2022 – 2026

JCT College of Engineering and Technology, Anna University

CGPA: 8.47 / 10

CERTIFICATIONS

- **Oracle Cloud Infrastructure 2023 —** AI Foundations Associate
- **Google AI Essentials**
- **IBM Data Science —** Selected Coursework (Python, Data Analysis, Machine Learning)

LANGUAGES

English · Malayalam · Tamil